Course Name : Community Engagement Project
Course Code: BIT25437A0B

Project Title: Inclusive Crowdsourcing: A Multilingual, SMS/Voice-Based Framework for Bridging Accessibility Gaps in Disaster Data Collection

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Introduction:

- Disaster response increasingly relies on **crowdsourced data** — citizen reports, social media and volunteered geographic information (VGI).
- Most existing systems are built around **English, Twitter-centric, smartphone** inputs.
- Yet many residents of flood-prone areas speak **Marathi or Hindi**, use **basic phones**, or have low digital literacy — so they go unheard.
- The most vulnerable (elderly, rural, low-income) are often the **least represented** in the very data meant to help them.
- Our project builds an **inclusive, multilingual, SMS/voice-ready** reporting framework, implemented as the working prototype **AwaazSetu** for Pune.

Literature Survey & Findings : (Paper 1 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
1.	Javed et al., 2024 — IndicVoices: Towards building an Inclusive Multilingual Speech Dataset for Indian Languages https://aclanthology.org/2024.findings-acl.639/	Year: 2024 Findings of ACL 2024 (Association for Computational Linguistics), pp. 10740–10782 Dataset / technology paper	• 7,348 hours of natural speech: read (9%), extempore (74%) and conversational (17%) • 16,237 speakers from 145 Indian districts covering 22 languages • 1,639 hours already transcribed, a median of 73 hours per language • Releases IndicASR, a speech recognition model supporting all 22 languages

Dataset(s) used: IndicVoices corpus — 7,348 h, 22 languages, 16,237 speakers, released by AI4Bharat (huggingface.co/datasets/ai4bharat/IndicVoices · arXiv:2403.01926).

Use in AwaazSetu (M1 – Intake): an open Indian-language speech model for transcribing Marathi and Hindi voice notes.

Literature Survey & Findings : (Paper 2 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
2.	Dursunoğlu et al., 2026 — Designing a Human-in-the-Loop AI System for Incident Consolidation and Severity-Aware Triage in Emergency Call Centers https://doi.org/10.59297/g2fhn369	Year: 2026 Proceedings of the International ISCRAM Conference, Vol. 23 Technology paper	• Full automation is unsafe in high-stakes emergencies; operators stay in control • Pipeline: speech-to-text → NLP → probabilistic severity → spatial-semantic clustering • Merges duplicate reports; adds real-time geospatial awareness • Synthetic surge test: good classification, calibrated confidence

Dataset(s) used: a synthetic, operationally grounded emergency-call dataset built by the authors (call surges, ambiguity, redundant reports); not publicly released.

Use in AwaazSetu (M2 – Triage): explained severity scoring, duplicate merging and coordinator verification of every incident.

Literature Survey & Findings : (Paper 3 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
3.	Hu et al., 2023 — Geo-knowledge-guided GPT models improve the extraction of location descriptions from disaster-related social media messages https://doi.org/10.1080/13658816.2023.2266495	Year: 2023 International Journal of Geographical Information Science (Taylor & Francis) Technology paper	• Disaster location descriptions go beyond place names and are hard for typical NER tools • Fuses geo-knowledge with GPT models using only 22 training examples • Over 40% improvement over NER approaches on Hurricane Harvey tweets • Geo-knowledge is indispensable for guiding GPT behaviour

Dataset(s) used: Hurricane Harvey (2017) tweets from the authors' annotated corpus; source: UNT Libraries Harvey Twitter collection · github.com/geoai-lab/HowDoPeopleDescribeLocations.

Use in AwaazSetu (M3 – Location): an India-specific few-shot extraction prompt combined with the Pune gazetteer of 450 places.

Literature Survey & Findings : (Paper 4 of 4)

Sr. No	Downloaded Paper title with link	Details of paper (year of publication /Journal name)	Findings of Paper (Min 4 in bullet points)
4.	Wang et al., 2025 — Mining socioenvironmental drivers of bias in social media rescue data for fairness-aware modeling: A case study of Hurricane Harvey https://doi.org/10.1016/j.ijdrr.2025.105717	Year: 2025 International Journal of Disaster Risk Reduction, Vol. 128 (Elsevier) Research paper	• Social media users do not represent the whole population, biasing rescue models • Defines Expected Rescue Requests and a Rescue Request Difference (RRD) index • 35 geographic, socioeconomic and digital-access variables; 1,807 requests, 534 block groups • % Minority and road density identified as sensitive attributes for fairness

Dataset(s) used: Hurricane Harvey rescue-request tweets (1,807 requests, 534 block groups) + 35 variables from ACS 2015 5-year estimates and CDC/ATSDR SVI, Houston MSA.

Use in AwaazSetu (M4 – Coverage): a coverage gap index per ward comparing observed reports with an expected baseline.

Project Title:

Inclusive Crowdsourcing: A Multilingual, SMS/Voice-Based Framework for Bridging Accessibility Gaps in Disaster Data Collection

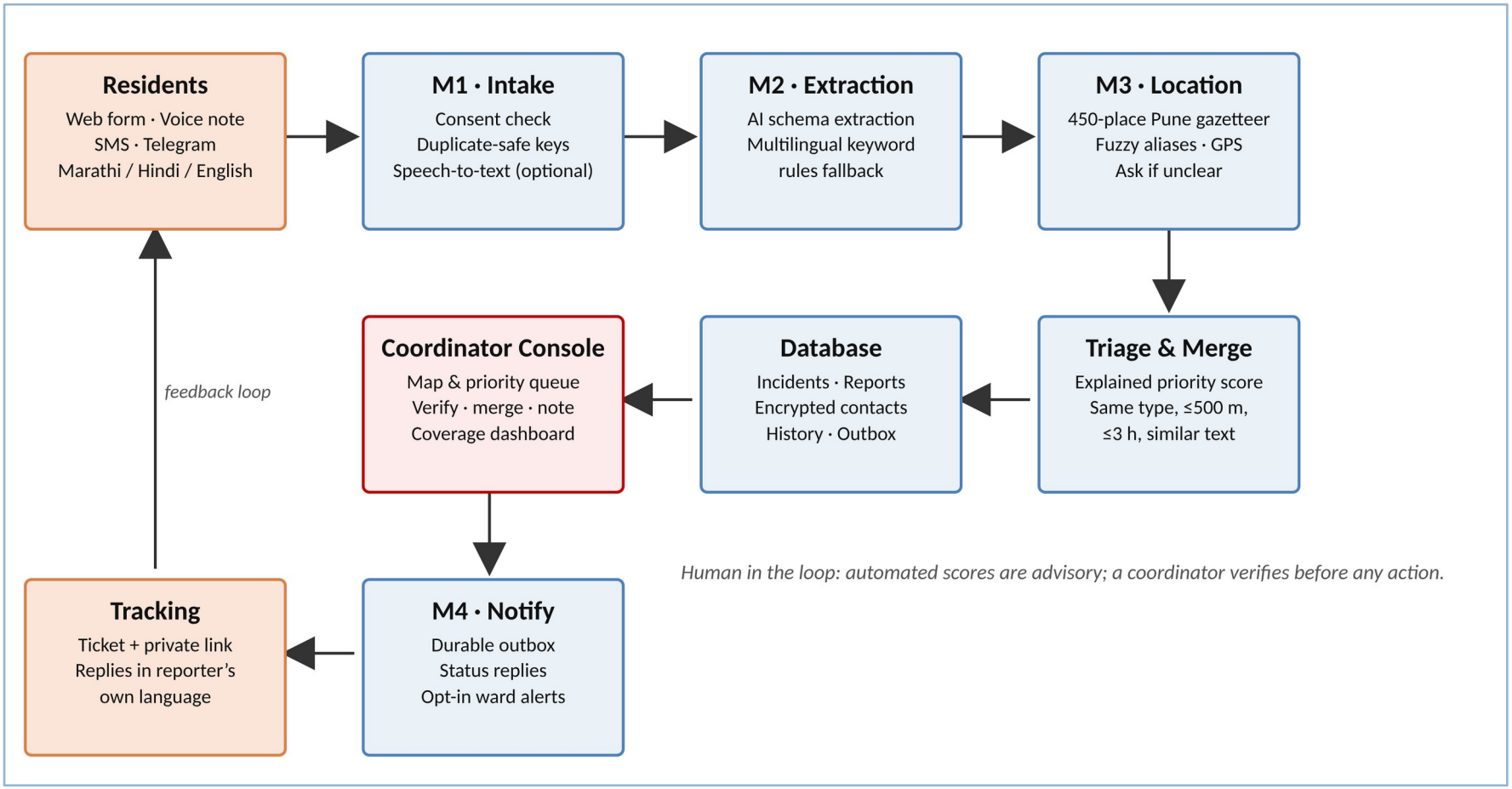
Working prototype: **AwaazSetu** (आवाज़ सेतु, "voice bridge") — a multilingual flood-reporting and coordination system for Pune community drills.

- Residents report by **web, voice, SMS or Telegram** in Marathi, Hindi or English.
- Coordinators see a **mapped, prioritised and explained** incident queue.
- A **coverage dashboard** shows which languages and areas are under-reported.

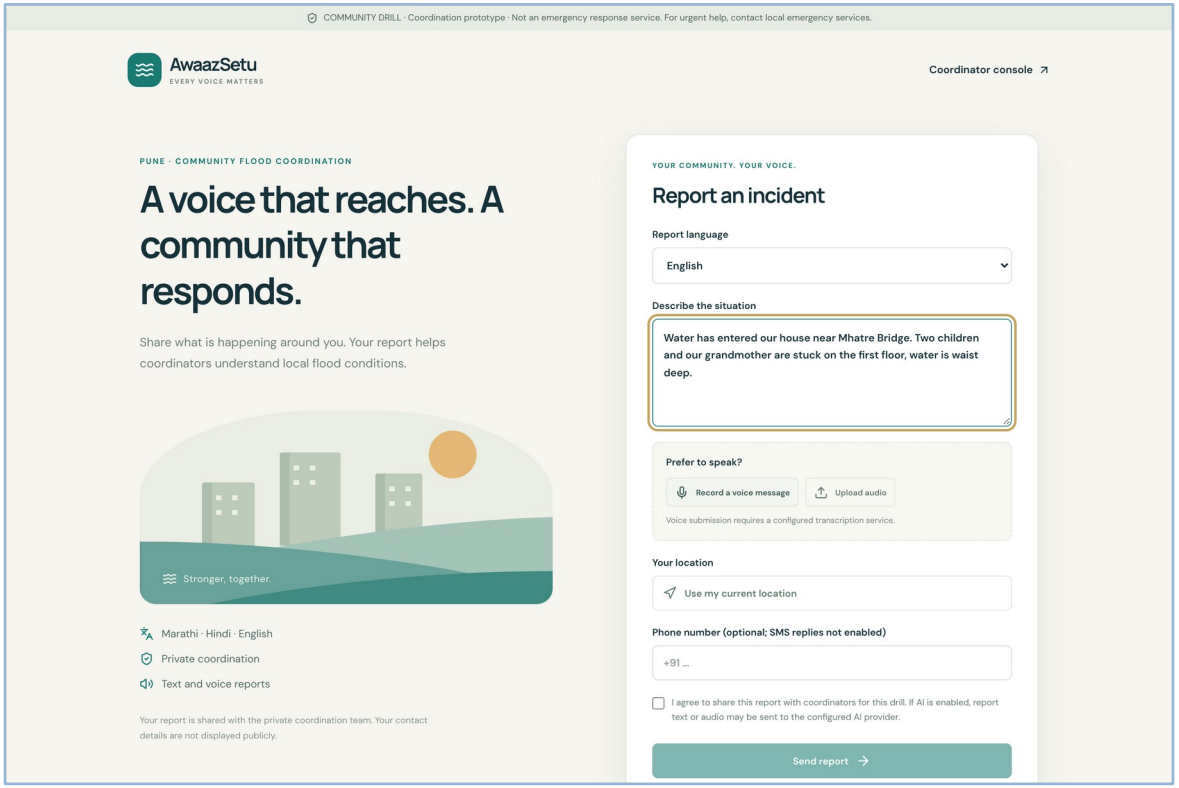
Objectives:

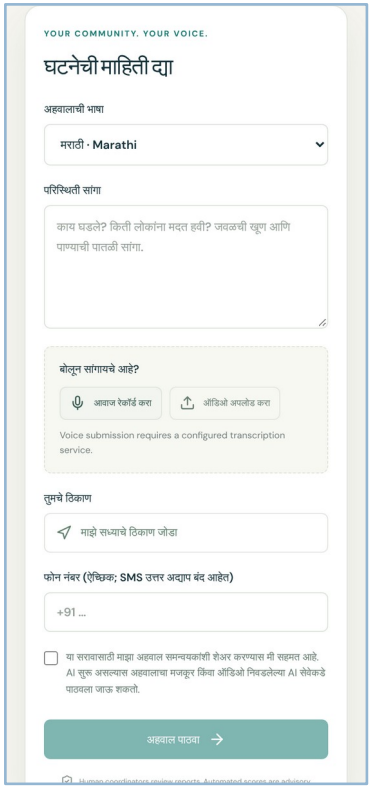
- Accept flood reports in **Marathi, Hindi and English** through **web, voice, SMS and Telegram**, so basic-phone users are included.
- Extract structured facts (incident type, people, water level, vulnerability) using **AI with a rules fallback** that works offline.
- Resolve informal **landmark names** to locations, and **ask instead of guessing** when unclear.
- Give each report a **transparent, explained priority score** and merge duplicates conservatively.
- Keep reporters informed in **their own language** and support **opt-in area alerts**.
- Measure **inclusion**: show which languages, channels and areas are under-represented.
- Protect **privacy** with consent, encrypted contacts and human review of every decision.

Proposed Methodology : (Block Diagram)



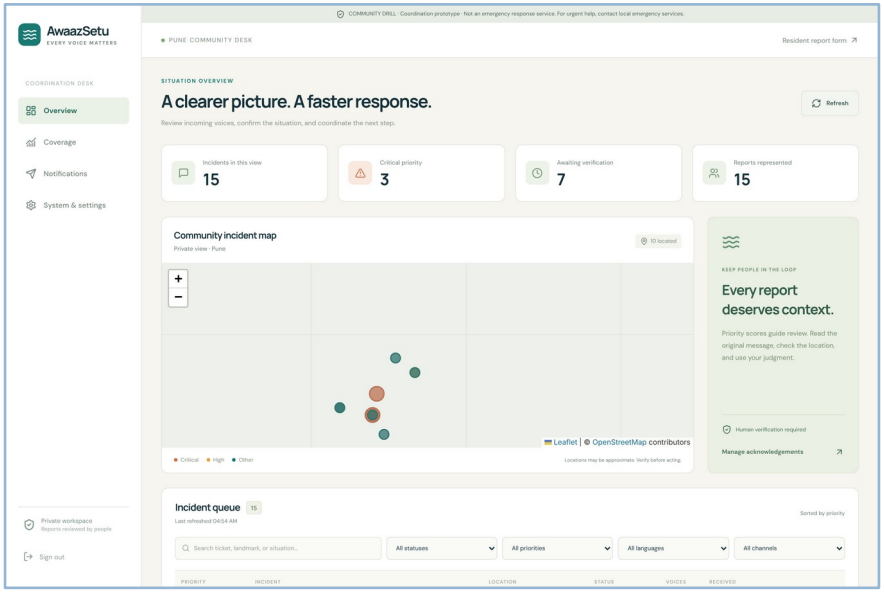
Implementation : Resident Interface

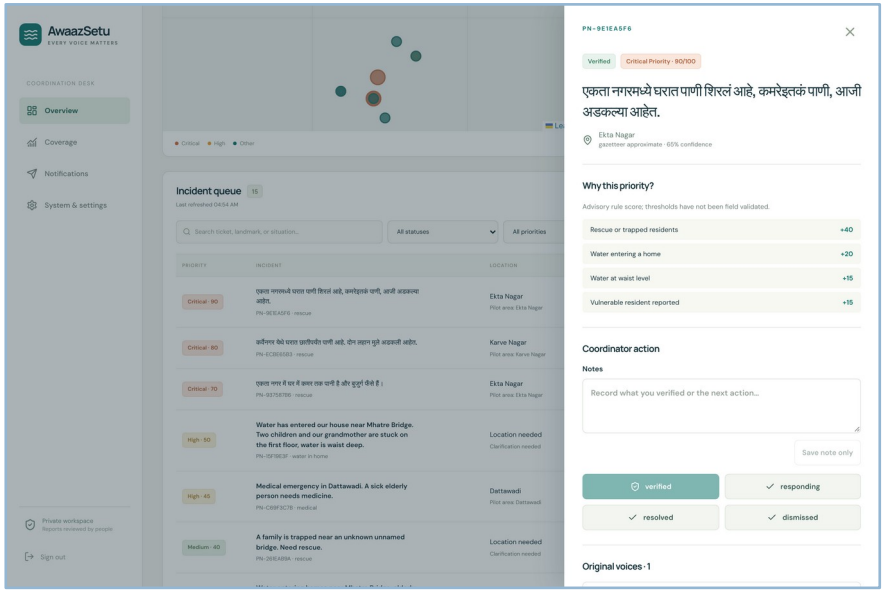




Features: three languages · voice with editable transcript · optional GPS · consent · ticket ID and private tracking link

Implementation : Coordinator Console





- **Overview:** live priority queue, private incident map, filters by status, priority, language and channel.
- **Incident detail:** score reasons, original message, verify / respond / resolve / dismiss, notes and merge with a reason.
- **Stack:** React + Vite, Leaflet, Recharts · FastAPI, SQLAlchemy, SQLite · optional OpenAI extraction and speech.

Result Analysis

Test / metric	Result	Notes
Backend automated tests	88 / 88 passed	pytest; API, triage, location, delivery, voice
Browser UI tests	10 / 10 passed	Playwright; resident and coordinator flows
Synthetic triage suite	150 / 150 exact match	50 cases each in EN, HI, MR; 7 fields
Report intake latency	median 7.4 ms, p95 9.8 ms	200 local reports, offline rules mode
Location lookup	0.24 ms per query	450-place Pune gazetteer
Field validation	Pending	Marathi speech accuracy, duplicate F1, landmark accuracy, SMS latency

Analysis: the rules engine handles all three languages consistently and responds in milliseconds, so it works offline and without AI cost. The synthetic suite is a regression check, **not field accuracy**; real accuracy needs a supervised community drill.

Conclusion:

- AwaazSetu shows that an **inclusive, multilingual, SMS- and voice-ready** channel can turn scattered flood reports into a **reviewed, located and prioritised** picture.
- Explained scores, conservative merging and **ask-don't-guess** locations keep coordinators in control.
- The coverage dashboard makes visible **who is still missing** from the data — the core gap found in the literature.
- **Future work:** connect the Android SMS gateway and Telegram bot, run a supervised drill, measure Marathi speech accuracy and duplicate F1, and add phone-ownership verification.

Thank You